

Individual Project Proposal (10%)

Project Title: OfferPacket Agent: A Multi-Agent, Tool-Using System for High-Signal Job Applications

1. Problem Statement & Motivation

Applying for jobs is a repetitive, high-effort process that often produces low-quality, generic application materials. Applicants spend significant time tailoring resumes and cover letters, yet recruiters still receive many applications that fail to clearly demonstrate role-specific fit. This creates inefficiency on both sides of the labor market.

The real-world business problem is twofold: applicants waste time producing low-signal artifacts, while recruiters and hiring managers spend time screening applications that do not clearly communicate value. Even strong candidates are often filtered out due to poor articulation of their skills rather than lack of ability.

The target users for this system are undergraduate and early-career applicants applying to internships or entry-level roles. Secondary users include career advisors or coaches who support multiple applicants. Recruiters and hiring managers are indirect beneficiaries through improved signal quality.

An agentic AI solution is appropriate because the task involves a structured, repeatable workflow rather than a single response. The system must analyze job requirements, map them to user evidence, generate tailored artifacts, review quality, and manage outputs. This requires planning, tool use, and coordination across specialized agents rather than a static chatbot interaction.

2. Use Cases & Scenarios

Scenario 1: Create a role-specific offer packet.

A user provides a job posting and selects relevant projects or experiences. The system extracts role requirements, maps them to evidence, and generates a one-page Offer Packet that clearly communicates value, proof-linked claims, and a short first-30-days plan.

Scenario 2: Generate recruiter outreach messages.

A user requests concise outreach and follow-up messages. The system drafts role- and company-specific messages that reference the generated Offer Packet.

Scenario 3: Track applications and summarize progress.

As the user applies to multiple roles, the system logs each application and generates summaries highlighting patterns, gaps, and next steps.

3. Initial System Design

Agents and Roles:

- Orchestrator/Planner: Coordinates the workflow and sequencing of tasks.
- Job Post Analyzer: Extracts responsibilities, requirements, and keywords.
- Evidence Mapper: Links job requirements to user-provided projects and proof.
- Offer Packet Writer: Generates the structured one-page Offer Packet.
- Critic/Reviewer: Checks clarity, evidence support, and formatting consistency.
- Tracker/Logger: Saves outputs and logs application metadata.

The system will implement a multi-agent workflow by Milestone II, demonstrating coordination among specialized agents rather than a single-agent solution.

Expected Tools and Data Sources:

The minimum viable system will use local file tools to export documents (PDF or DOCX) and store tracking data in a local CSV or JSON file. User-provided resume text, portfolio links, and job posting content will serve as inputs. Optional integrations such as Google Drive or Google Sheets may be explored if time permits.

User-Facing Interface:

A simple front-end interface will allow users to input job postings and experience, trigger artifact generation, preview outputs, download files, and view a basic application tracker. Streamlit or a lightweight web interface will be used depending on feasibility.

4. Feasibility & Risks

Potential challenges include hallucinated or unsupported claims, tool failures during export or saving, long-context inputs, and privacy concerns. These risks will be mitigated through an evidence-mapping step, a critic agent that enforces quality standards, fallback output modes, and local-only data handling.

The primary constraint is the limited project duration. The scope will prioritize a stable, well-evaluated MVP over complex external integrations.

5. Scaled Timeline (4–5 Weeks)

Week 1: Finalize scope, agent roles, and output formats. Set up repository and basic UI skeleton.

Week 2 (Milestone I – Front-End): Implement user input interface and basic single-agent draft generation with on-screen output.

Week 3: Add multi-agent workflow, document export, and local application tracking.

Week 4 (Milestone II): Demonstrate full multi-agent pipeline with planner, writer, and critic. Improve robustness.

Week 5 (Milestone III & Finalization): Conduct pilot evaluation, analyze results, refine system, and complete final report. A brief recorded walkthrough or presentation may be prepared if required.

6. Evaluation Plan

The system will be evaluated using a small pilot set of real job postings. Behavior metrics will include requirement coverage, evidence support rate, critic-detected issues, and tool success rate. Usefulness will be assessed through user ratings on clarity, confidence to submit, and time saved, along with short qualitative feedback.

This evaluation will demonstrate both technical correctness and practical value of the agentic system.

Deliverable

This proposal is submitted as a PDF document formatted to 2–3 pages, double-spaced, with one-inch margins on U.S. letter paper.